



Bacterial blight

What it does

Bacterial blight is caused by *Xanthomonas oryzae* pv. *oryzae*.

It causes wilting of seedlings and yellowing and drying of leaves.

Why and where it occurs

The disease is most likely to develop in areas that have weeds and stubbles of infected plants. It can occur in both tropical and temperate environments, particularly in irrigated and rainfed lowland areas. In general, the disease favors temperatures at 25–34°C, with relative humidity above 70%.



It is commonly observed when strong winds and continuous heavy rains occur, allowing the disease-causing bacteria to easily spread through ooze droplets on lesions of infected plants.

Bacterial blight can be severe in susceptible rice varieties under high nitrogen fertilization.

How to identify

Check for wilting and yellowing of leaves, or wilting of seedlings (also called *kresek*).

On seedlings, infected leaves turn grayish green and roll up. As the disease progresses, the leaves turn yellow to straw-colored and wilt, leading whole seedlings to dry up and die.

Kresek on seedlings may sometimes be confused with early rice stem borer damage.

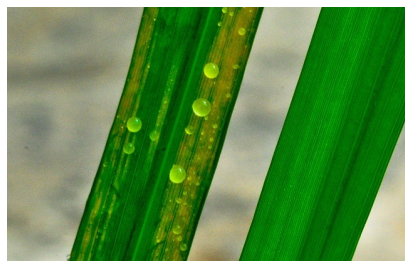
To distinguish kresek symptoms from stem borer damage, squeeze the lower end of infected seedlings between the fingers. Kresek symptoms should show yellowish bacterial ooze coming out of the cut ends. Unlike plants infested with stem borer, rice plants with kresek are not easily pulled out from soil.



Lesions caused by bacterial blight



- On older plants, lesions usually develop as water-soaked to yellow-orange stripes on leaf blades or leaf tips or on mechanically injured parts of leaves. Lesions have a wavy margin and progress toward the leaf base.
- On young lesions, bacterial ooze resembling a milky dew drop can be observed early in the morning. The bacterial ooze later on dries up and becomes small yellowish beads underneath the leaf.



Bacterial ooze



Dried up bacterial ooze

- Old lesions turn yellow to grayish white with black dots due to the growth of various saprophytic fungi. On severely infected leaves, lesions may extend to the leaf sheath.

To quickly diagnose bacterial blight on leaf:

- cut a young lesion across and place in a transparent glass container with clear water
- after a few minutes, hold the container against light and observe for thick or turbid liquid coming from the cut end of the leaf

Why is it important

Bacterial blight is one of the most serious diseases of rice. The earlier the disease occurs, the higher the yield loss.

Yield loss due to bacterial blight can be as much as 70% when susceptible varieties are grown, in environments favorable to the disease.

When plants are infected at booting stage, bacterial blight does not affect yield but results in poor quality grains and a high proportion of broken kernels.

How to manage

Planting resistant varieties has been proven to be the most efficient, most reliable, and cheapest way to control bacterial blight.

Other disease control options include:

- Use balanced amounts of plant nutrients, especially nitrogen.
- Ensure good drainage of fields (in conventionally flooded crops) and nurseries.
- Keep fields clean. Remove weed hosts and plow under rice stubble, straw, rice ratoons and volunteer seedlings, which can serve as hosts of bacteria.
- Allow fallow fields to dry in order to suppress disease agents in the soil and plant residues.

Content experts: CM Vera Cruz (email: c.veracruz@irri.org), I Ona, NP Castilla, and R Oplencia



Bacterial leaf streak

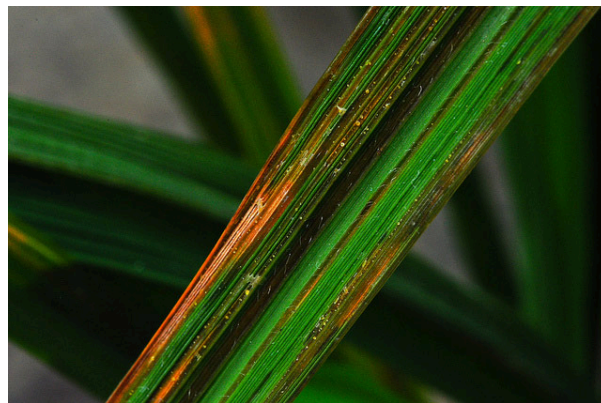
What it does

Bacterial leaf streak is caused by *Xanthomonas oryzae* pv. *oryzicola*.

Infected plants show browning and drying of leaves. Under severe conditions, this could lead to reduced grain weight due to loss of photosynthetic area.

Why and where it occurs

Bacterial leaf streak occurs in areas with high temperature and high humidity.



It is transmitted through seeds and infected stubbles to the next planting season. It can occur in fields where *X. oryzae* pv. *oryzicola* bacteria is present on leaves, in the water, or in the debris left after harvest.

Particularly, the disease is common in tropical and subtropical regions of Asia, Africa (including Madagascar), South America, and Australia.

It can affect the plant during early stages, from maximum tillering to panicle initiation. Mature rice plants can easily recover from leaf streak and have minimal grain yield losses.

How to identify

Check for lesions:

- Symptoms initially appear as small, water-soaked, linear lesions between leaf veins. These streaks are initially dark green and later become light brown to yellowish gray.
- The lesions are translucent when held against the light.
- Entire leaves may become brown and die when the disease is very severe.
- Under humid conditions, yellow droplets of bacterial ooze, which contain masses of bacterial cells, may be observed on the surface of leaves.





Linear lesions

Enlarged lesions

Water-soaked lesions

Bacterial leaf streak may be confused with narrow brown spot.

To confirm:

- leaf streak lesions are usually thinner than those of narrow brown spot
- narrow brown spot lesions are not translucent, nor do they produce bacterial ooze

When the advancing part of the streaks are cut and placed in a glass with water, a mass of bacterial cells can usually be seen oozing out of the leaf, which makes the water turbid after five minutes.

Bacterial leaf streak may also be confused with bacterial blight.

Blight and streak symptoms look the same at the early stage of infection, particularly if lesions of both diseases occur on the same leaf. Tests (i.e., biochemical, pathogenicity, or serological) can be done to identify the disease.

Why is it important

Based on reported cases, yield loss caused by bacterial leaf streak can range from 8–17% in the wet season, and 1–3 % in the dry season.

How to manage

To prevent and effectively manage bacterial leaf streak:

- Plant resistant varieties.
- Treat seeds with hot water.
- Keep fields clean—remove weed hosts and plow under rice stubble, straw, rice ratoons, and volunteer seedlings, which may be infected by the bacteria.
- Use balanced amounts of plant nutrients, especially nitrogen.
- Ensure good drainage of fields (in conventionally flooded crops) and nurseries.
- Drain the field during severe flood.
- Dry the field during the fallow period to kill the bacteria in the soil and in plant residues.
- In cases of severe infection, when yield may be affected, a copper-based fungicide applied at heading can be effective in controlling the disease.

Content experts: Adam Sparks (email: a.sparks@irri.org), NP Castilla, and CM Vera Cruz



Blast (leaf and collar)

What it does

Blast is caused by the fungus *Magnaporthe oryzae*. It can affect all above ground parts of a rice plant: leaf, collar, node, neck, parts of panicle, and sometimes leaf sheath.

Why and where it occurs

Blast can occur wherever blast spores are present.

It occurs in areas with low soil moisture, frequent and prolonged periods of rain shower, and cool temperature in the daytime. In upland rice, large day-night temperature differences that cause dew formation on leaves and overall cooler temperatures favor the development of the disease.

Rice can have blast in all growth stages. However, leaf blast incidence tends to lessen as plants mature and develop adult plant resistance to the disease.



How to identify

Check the leaf and collar for lesions:

- Initial symptoms appear as white to gray-green lesions or spots, with dark green borders.
- Older lesions on the leaves are elliptical or spindle-shaped and whitish to gray centers with red to brownish or necrotic border.

Check for other symptoms:

- Some resemble diamond shape, wide in the center and pointed toward either ends.
- Lesions can enlarge and coalesce, growing together, to kill the entire leaves.



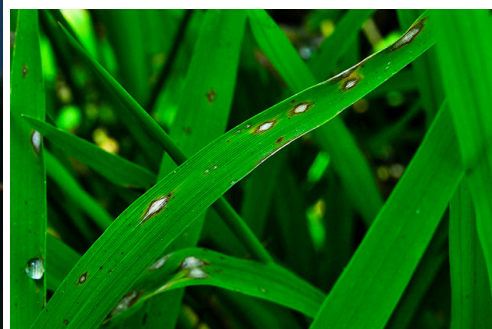
Collar blast



Old lesions



Spindle-shaped lesions



Diamond-shaped lesions

Blast lesions can commonly be confused with Brown Spot lesions.

Leaf blast lesions are usually elongated and pointed at each end, while brown spot lesions tend to be more round, brown in color and have a yellow halo surrounding the lesion.



Why is it important

Rice blast is one of the most destructive diseases of rice. A leaf blast infection can kill seedlings or plants up to the tillering stage. At later growth stages, a severe leaf blast infection reduces leaf area for grain fill, reducing grain yield.

Leaf blast can kill rice plants at seedling stage and cause yield losses in cases of severe infection.

How to manage

The primary control option for blast is to plant resistant varieties. Contact your local agriculture office for up-to-date lists of varieties available.

Other crop management measures can also be done, such as:

- Adjust planting time. Sow seeds early, when possible, after the onset of the rainy season.
- Split nitrogen fertilizer application in two or more treatments. Excessive use of fertilizer can increase blast intensity.
- Flood the field as often as possible.

Silicon fertilizers (e.g., calcium silicate) can be applied to soils that are silicon deficient to reduce blast. However, because of its high cost, silicon should be applied efficiently. Cheap sources of silicon, such as straws of rice genotypes with high silicon content, can be an alternative. Care should be taken to ensure that the straw is free from blast as the fungus can survive on rice straw and the use of infected straw as a silicon source can spread the disease further.

Systemic fungicides like triazoles and strobilurins can be used judiciously for control to control blast. A fungicide application at heading can be effective in controlling the disease.

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Blast (node and neck)

What it does

Blast is caused by the fungus *Magnaporthe oryzae*. It can affect all above ground parts of a rice plant: leaf, collar, node, neck, parts of panicle, and sometimes leaf sheath.

Node and neck blast commonly occur together and have similar symptoms.

When a node or neck blast infection is present, it can cause plants to develop few or no grains at all.

Why and where it occurs

Blast can occur wherever blast spores are present.

It occurs in areas with low soil moisture, frequent and prolonged periods of rain shower, and cool temperature in the daytime. In upland rice, large day-night temperature differences that cause dew formation on leaves and overall cooler temperatures favor the development of the disease.

Rice can have blast in all growth stages. However, leaf blast incidence tends to lessen as plants mature and develop adult plant resistance to the disease.



How to identify

Check the nodes and necks for lesions and rotting:

- Node infection occurs in banded pattern.
- Lesions on the node are blackish to grayish brown.
- Infected nodes can cause the culm or the part of the plant that holds the panicle to break.
- Lesions on the neck are grayish brown and can cause girdling, making the neck and the panicle fall over.
- If infection of the neck occurs before milky stage, no grain is formed, but if infection occurs later, poor quality grains are formed.





- Neck and node blast can also cause whiteheads or white panicles, similar to stem borer infection.
- Whiteheads caused by stem borers can be pulled apart from the plant, the stem will separate at the point where the insect bored into it.
- With neck and node blast, tugging on the stem will not result in removal.

Why is it important

Rice blast is one of the most destructive diseases of rice. Because neck and node blast affect the panicle directly, yield losses in affected plants are very high.

How to manage

The primary control option for blast is to plant resistant varieties. Contact your local agriculture office for up-to-date lists of varieties available.

Other crop management measures can also be done, such as:

- Adjust planting time. Sow seeds early, when possible, after the onset of the rainy season.
- Split nitrogen fertilizer application in two or more treatments. Excessive use of fertilizer can increase blast intensity.
- Flood the field as often as possible.

Silicon fertilizers (e.g., calcium silicate) can be applied to soils that are silicon deficient to reduce blast. However, because of its high cost, silicon should be applied efficiently. Cheap sources of silicon, such as straws of rice genotypes with high silicon content, can be an alternative.

Care should be taken to ensure that the straw is free from blast as the fungus can survive on rice straw and the use of infected straw as a silicon source can spread the disease further.

Systemic fungicides like triazoles and strobilurins can be used judiciously for control to control blast. A fungicide application at heading can be effective in controlling the disease in cases where the risk of yield losses is high.

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Brown spot

Brown spot has been historically largely ignored as one of the most common and most damaging rice diseases.

What it does

Brown spot is a fungal disease that infects the coleoptile, leaves, leaf sheath, panicle branches, glumes, and spikelets.

Its most observable damage is the numerous big spots on the leaves which can kill the whole leaf. When infection occurs in the seed, unfilled grains or spotted or discolored seeds are formed.

Why and where it occurs

The disease can develop in areas with high relative humidity (86–100%) and temperature between 16 and 36°C. It is common in unflooded and nutrient-deficient soil, or in soils that accumulate toxic substances.

For infection to occur, the leaves must be wet for 8–24 hours.

The fungus can survive in the seed for more than four years and can spread from plant to plant through air. Major sources of brown spot in the field include:

- infected seed, which give rise to infected seedlings
- volunteer rice
- infected rice debris
- weeds

Brown spot can occur at all crop stages, but the infection is most critical during maximum tillering up to the ripening stages of the crop.

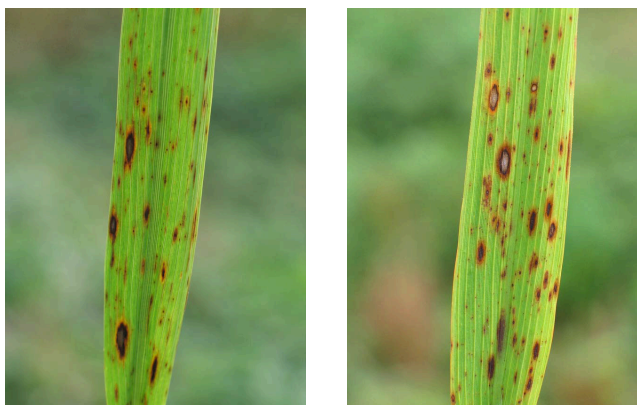
How to identify

Check for lesions:

- Infected seedlings have small, circular, yellow brown or brown lesions that may girdle the coleoptile and distort primary and secondary leaves.
- Starting at tillering stage, lesions can be observed on the leaves. They are initially small, circular, and dark brown to purple-brown.
- Fully developed lesions are circular to oval with a light brown to gray center, surrounded by a reddish brown margin caused by the toxin produced by the fungi.

On susceptible varieties, lesions are 5–14 mm long which can cause leaves to wilt. On resistant varieties, the lesions are brown and pinhead-sized.





Spots and lesions on leaves

Lesions on leaf sheaths are similar to those on the leaves. Infected glumes and panicle branches have dark brown to black oval spots or discoloration on the entire surface.

Spikelets can also be infected. Infection of florets leads to incomplete or disrupted grain filling and a reduction in grain quality. The disease-causing fungi can also penetrate grains, causing '*pecky rice*', a term used to describe spotting and discoloration of grains.

In certain rice varieties, brown spot lesions can be mistaken for blast lesions. To confirm, check if spots are circular, brownish, and have a gray center surrounded by a reddish margin.

Why is it important

Brown spot causes both quantity and quality losses.

On average, the disease causes 5% yield loss across all lowland rice production in South and Southeast Asia. Severely infected field can have as high as 45% yield loss.

Heavily infected seeds cause seedling blight and lead to 10–58% seedling mortality. It also affects the quality and the number of grains per panicle, and reduces the kernel weight.

In terms of history, Brown spot was considered to be the major factor contributing to the [Great Bengal Famine](#) in 1943.

How to manage

Improving soil fertility is the first step in managing brown spot. To do this:

- monitor soil nutrients regularly
- apply required fertilizers
- for soils that are low in silicon, apply calcium silicate slag before planting

Fertilizers, however, can be costly and may take many cropping seasons before becoming effective. More economical management options include:

- Use resistant varieties.
Contact your local agriculture office for up-to-date lists of varieties available.
- Use fungicides (e.g., iprodione, propiconazole, azoxystrobin, trifloxystrobin, and carbendazim) as seed treatments.
- Treat seeds with hot water (53–54°C) for 10–12 minutes before planting, to control primary infection



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Fact sheet

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False smut

What it does

False smut causes chalkiness of grains which leads to reduction in grain weight. It also reduces seed germination.

Why and where it occurs

The disease can occur in areas with high relative humidity (>90%) and temperature ranging from 25–35 °C.

Rain, high humidity, and soils with high nitrogen content also favors disease development. Wind can spread the fungal spores from plant to plant.

False smut is visible only after panicle exsertion. It can infect the plant during flowering stage.

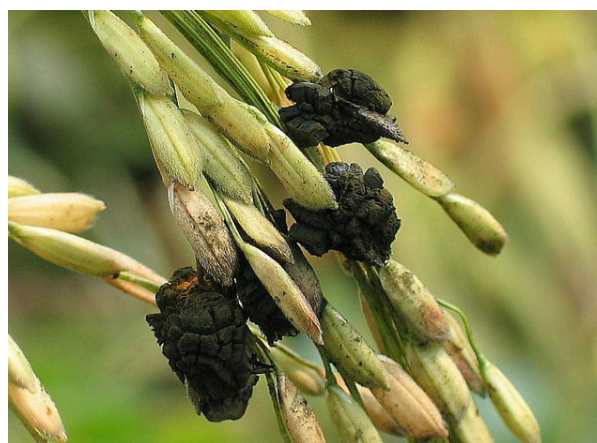
How to identify

Check for presence of velvety smut balls on spikelets.

Plants infected with false smut have individual rice grain transformed into a mass of spore balls. These spore balls are initially orange, and then turn into greenish black when these mature.

In most cases, not all spikelets of a panicle are affected, but spikelets neighboring smut balls are often unfilled.

- Individual rice grain transformed into a mass of velvety spores or yellow fruiting bodies
- Growth of velvety spores enclose floral parts
- Immature spores slightly flattened, smooth, yellow, and covered by a membrane
- Growth of spores result to broken membrane
- Mature spores orange and turn yellowish green or greenish black
- Only few grains in a panicle are usually infected and the rest are normal



Spore balls are initially orange and turn greenish black when mature

Why is it important

The disease affects the early flowering stage of the rice crop when the ovary is destroyed. The second stage of infection occurs when the spikelet nearly reaches maturity.

These causes chalkiness and can reduce 1,000-grain weight. It also causes a reduction in seed germination of up to 35%. In damp weather, the disease can be severe and losses can reach 25%. In India, a yield loss of 7–75% was observed.



How to manage

- Keep the field clean.
- Remove infected seeds, panicles, and plant debris after harvest.
- Reduce humidity levels through alternate wetting and drying (AWD) rather than permanently flooding the fields.
- Where possible, perform conservation tillage and continuous rice cropping.
- Use moderate rates of Nitrogen.
- Use certified seeds.
- Resistant varieties have been reported. Contact your local agriculture office for an up-to-date list of available varieties.
- Treat seeds at 52°C for 10 min.

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Leaf Scald

What it does

Leaf scald is a fungal disease caused by *Microdochium oryzae*, which causes the scalded appearance of leaves.

Why and where it occurs

Disease development usually occurs late in the season on mature leaves and is favored by wet weather, high nitrogen fertilization, and close spacing.

It develops faster in wounded than in unwounded leaves.

The sources of infection are seeds and crop stubbles. Wet weather and high doses of nitrogenous fertilizer favor the disease.

Leaf scald commonly occurs in Central and South America, where it has caused significant yield losses. It also occurs in Asia, Africa, and the U.S. The disease is found in upland, rainfed, irrigated, and mangrove areas.

How to identify

Check the plant for the following symptoms:

- zonate lesions of alternating light tan and dark brown starting from leaf tips or edges
- oblong lesions with light brown halos in mature leaves
- translucent leaf tips and margins

Individual lesions are 1–5 cm long and 0.5–1 cm wide or may almost cover the entire leaf. The continuous enlargement and coalescing of lesions result in blighting of a large part of the leaf blade. The affected areas dry out giving the leaf a scalded appearance.

In some countries, lesions rarely develop the zonate pattern and only the scalding symptom is evident.

In Costa Rica, the disease has been reported to cause decay of coleoptiles, with red brown infection, root rot, and a head blight that caused considerable sterility, flower deformation and glume discoloration.

Infected leaf tips also split near the midrib especially when there are strong winds.

Leaf scald can be confused with leaf blight. To confirm scald, visually examine the leaf for scalded appearance. Immerse cut leaves in clear water for 5–10 minutes, if no ooze comes out, then it is leaf scald.

Why is it important

Leaf scald has caused considerable losses in Latin America and West Africa.





- Use resistant varieties.
Contact your local agriculture office for an up-to-date list of available varieties.
- Avoid high use of fertilizer. Apply Nitrogen in split.
- Use benomyl, carbendazim, quitozene, and thiophanate-methyl to treat seeds.
- In the field, spraying of benomyl, fentin acetate, edifenphos, and validamycin significantly reduce the incidence of leaf scald. Foliar application of captafol, mancozeb, and copper oxychloride also reduces the incidence and severity of the fungal disease.

To prevent pathogen survival across cropping seasons:

- Remove weeds.
- Plow under of rice stubbles.
- Remove infected rice ratoons.

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Sheath blight

What it does

Sheath blight is a fungal disease caused by *Rhizoctonia solani*. Infected leaves senesce or dry out and die more rapidly, young tillers can also be destroyed.

As a result, the leaf area of the canopy can significantly be reduced by the disease. This reduction in leaf area, along with the diseased-induced senescence of leaves and young infected tillers are the primary causes of yield reduction.

Why and where it occurs

Sheath blight occurs in areas with high temperature (28–32°C), high levels of nitrogen fertilizer, and relative humidity of crop canopy from 85–100%.

Plants are more vulnerable to sheath blight during the rainy season.

High seeding rate or close plant spacing, dense canopy, disease in the soil, sclerotia or infection bodies floating on the water, and growing of high yielding improved varieties also favor disease development.

How to identify

Symptoms are usually observed from tillering to milk stage in a rice crop and include the following:

- oval or ellipsoidal greenish gray lesions, usually 1–3 cm long, on the leaf sheath, initially just above the soil or water level in the case of conventionally flooded rice.
- under favorable conditions, these initial lesions multiply and expand to the upper part of the sheaths, the leaves, and then spread to neighboring tillers belonging to different hills (transplanted rice) or plants (direct-seeded rice).
- lesions on the leaves usually have irregular lesions, often with gray-white centers and brown margins as they grow older.
- in subtropical environments, the disease is mostly initiated by sclerotia (up to two million of which can be produced per square meter in a diseased crop).

In tropical ecosystems, however, the role of sclerotia in initiating the disease is uncertain. In any case, the pathogen does not produce aerially dispersed spores; its dispersal is essentially dependent on hyphae running on plant tissues and producing new infections



Sheath blight has symptoms similar to stem rot and stem borer infestation. To confirm the cause of disease



later becomes grayish white with brown margin). Also check for presence of sclerotia.

Why is it important

Sheath blight is considered to be an important disease next to rice blast. Rice sheath blight is an increasing concern for rice production especially in intensified production systems.

In Japan, the disease has caused a yield loss of as high as 20% and affected about 120,000–190,000 hectares. A yield loss of 25% was reported if the flag leaves are infected. In the United States, a yield loss of 50% was reported when susceptible cultivars were planted. Sheath blight has also caused a yield loss of 6% in tropical Asia.

How to manage

There is currently no resistant rice variety available for cultivation. The main management options available to minimize sheath blight include:

- use a reasonable level of fertilizer adapted to the cropping season.
- use reasoned density of crop establishment (direct seeding or transplanting).
- carefully control of weeds, especially on the levees.
- drain rice fields relatively early in the cropping season to reduce sheath blight epidemics.
- use fungicide to treat seeds.
- improve canopy architecture by reducing seeding rate or providing wider plant spacing.

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Stem rot

What it does

Stem rot leads to formation of lesions and production of chalky grains and unfilled panicles.

Why and where it occurs

The infection bodies or sclerotia are found in the upper soil layer. They survive in air-dry soil, buried moist rice soil, and in tap water. They can also survive on straw, which is buried in the soil. The sclerotia float on irrigation water and infect newly planted rice during land preparation.

Infection is high on plants with wounds as a result of lodging or insect attack. The panicle moisture content and nitrogen fertilizer also influence disease development.



How to identify

Check the plant for the following symptoms:

Infected stem rots

- visible numerous tiny white and black sclerotia and mycelium inside the infected culms
- infected culm lodges and caused unfilled panicles and chalky grain

Initial symptoms are small, irregular black lesions on the outer leaf sheath near water level. Lesions expand as the disease advances.

Severe infection causes tiller death.

Why is it important

The infection is seen on the rice crop during early heading and grain filling. The leaf sheaths decay and cause lodging and lower grain filling. It can cause heavy losses in many countries.

For example, in Japan, there are 51,000–122,000 hectares infected and estimated annual losses of 16,000–35,000 due to this disease. In Vietnam, the Philippines, and India, losses from 30% to 80% were recorded.

How to manage

- Use resistant cultivars.
Contact your local agriculture office for an up-to-date list of available varieties.
- Burn straw and stubble or any crop residue after harvest or let the straw decompose.
- Drain the field to reduce sclerotia.
- Balance the use of fertilizer or perform split application with high potash and lime to increase soil pH.
- Chemicals such as fentin hydroxide sprayed at the mid-tillering stage, thiophanate-methyl sprayed at



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Tungro

What it does

Rice tungro disease is caused by the combination of two viruses, which are transmitted by leafhoppers. It causes leaf discoloration, stunted growth, reduced tiller numbers and sterile or partly filled grains.

Tungro infects cultivated rice, some wild rice relatives and other grassy weeds commonly found in rice paddies.

Why and where it occurs

Tungro disease viruses are transmitted from one plant to another by leafhoppers that feed on tungro-infected plants. The most efficient vector is the green leafhopper.

Leafhoppers can acquire the viruses from any part of the infected plant by feeding on it, even for a short time. It can, then, immediately transmit the viruses to other plants within 5–7 days. The viruses do not remain in the leafhopper's body unless it feeds again on an infected plant and re-acquires the viruses.

Tungro infection can occur during all growth stages of the rice plant. It is most frequently seen during the vegetative phase. Plants are most vulnerable at tillering stage.

Tungro incidence depends on the availability of the virus sources and vector population. Other than infected rice plants in the farmer's field, other primary sources for tungro, include:

- stubble of previous crops
- new growth from infected stubbles that had not been properly plowed under and harrowed effectively
- volunteer rice
- infected plants in nearby rice fields

Seedlings raised in nurseries or seedbeds can also be infected with Tungro prior to transplanting and can be a primary source of virus.

Transplanting seedlings from nurseries in tungro-infected areas has also shown to increase infection rates in the field, particularly, in cases where seedbed is in a tungro-endemic area or when the nursery duration is 5–6 weeks.

However, this is not believed to be a very strong mechanism in initiating epidemics, because the competitiveness of tungro-infected seedlings is low; they can die rapidly after transplanting.

Why is it important

Tungro is one of the most damaging and destructive diseases of rice in South and Southeast Asia. In severe cases, Tungro susceptible varieties infected at an early growth stage could have as high as 100% yield loss.

Once tungro is present in the field, it increases rapidly in young rice plants. Leafhopper vectors prefer to feed on young rice plants. They also acquire tungro viruses more efficiently from younger infected plants.

How to identify





Check leaves for discoloration.

Yellow or orange-yellow discoloration is noticeable in tungro-infected plants. Discoloration begins from the leaf tip and extends down to the blade or the lower leaf portion. Infected leaves may also show mottled or striped appearance, rust-colored spots, and inter-veinal necrosis.



Tungro-infected plant



Yellowing of leaves



Tungro-infected field



Green leafhopper

Tungro-infected plants also show symptoms of stunting, delayed flowering which may delay maturity, reduced number of tillers, small and not completely exerted panicles, as well as a higher than normal percentage of sterile panicles or partially filled grains, covered with dark brown blotches.

The degree of stunting and of leaf discoloration varies with rice varieties, strains of the viruses, the age of the plant when infected, and with the environment. In varieties that carry some resistance to the disease, infected plants exhibit no discoloration or only a mild discoloration that may disappear as the plants mature.

Tungro symptoms can be mistaken for physiological disorders.

Specifically,

- the yellowing of the plant and its stunted height can be confused as nitrogen and zinc deficiencies and water stress,
- pest infestation such as stem borer infestation, plant hopper infestation, and rat damage, and
- other diseases such as grassy stunt virus disease and orange leaf.

Where available, Tungro can be confirmed using serological tools such as Latex agglutination test, Enzyme Linked Immunosorbent Assays (ELISA) and Rapid Immunofilter Paper Assays (RIPA).

How to manage

Once a rice plant is infected by tungro, it cannot be cured.

Preventive measures are more effective for the control of tungro than direct disease control measures. Using insecticides to control leafhoppers is often not effective, because green leafhoppers continuously move to surrounding fields and spread tungro rapidly in very short feeding times.

The most practical measures at present, include

- Grow tungro or leafhopper resistant varieties.
This is the most economical means of managing the disease. There are tungro-resistant varieties available for the Philippines, Malaysia, Indonesia, India, and Bangladesh. Contact your local agriculture office for up-to-date lists of varieties available.
- Practice synchronous planting with surrounding farms.
Delayed or late planting, relative to the average date in a given area, makes the field susceptible for Tungro. Late-planted fields also pose a risk to early planting in the next season.

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- Plow infected stubbles immediately after harvest to reduce inoculum sources and destroy the eggs and breeding sites of green leaf hopper.

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